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EDITORIAL

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Machine learning in manufacturing and industry 4.0 applications

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ABSTRACT

Themachinelearning(ML)fieldhasdeeplyimpactedthemanufacturingindustryinthecontext oftheIndustry4.0paradigm.Theindustry4.0paradigmencouragestheusageofsmartsensors, devices, and machines, to enable smart factories that continuously collect data pertaining to pro-

Additionally, the ability of ML techniques to provide predictive insights has enabled discerning complex manufacturing patterns and offers a pathway for an intelligent decision support system in a varietyofmanufacturingtaskssuchasintelligentandcontinuousinspection,predictivemainte-nance, quality improvement, process optimisation, supply chain management, and task scheduling. WhiledifferentMLtechniqueshavebeenusedinavarietyofmanufacturingapplicationsinthe past, many open questions and challenges remain, from Big data curation, storage, and understand-ing, data reasoning to enable real-time actionable intelligence to topics such as edge computing andcybersecurityaspectsofsmartmanufacturing.Hence,thisspecialissueisfocusedonbring-**ing together a wide range of researchers to report the latest efforts in the fundamental theoretical** aswellasexperimentalaspectsofMLandtheirapplicationsinmanufacturingandproduction systems.

KEYWORDS

machine learning; industry 4.0; manufacturing analytics; big data in manufacturing; cloud manufacturing; fault detection

Introduction

Manufacturingcanbedefinedasthefabricationor assemblyofcomponentsintofinishedproductsona large scale (Encyclopaedia Britannica 2020). It is one of themostimportantindustriesintheworld'seconomy wherein, it accounts for approximately 16% of the global GDP in 2019 and has generated an output of 13.9 trillion globally(Sodani2021).Oneofthemostcriticalmanufacturing goals is to produce more high quality products atminimumcosts.Butmanufacturingproductscanbe a very expensive and complicated process for businesses thatdonothavetheassociatedresourcesandtoolsto designanddevelopqualityproducts(AMFGAI2019). Overthelastcoupleofcenturies,thehistoryofmanufacturinghaschangeddramatically.Insteadofitems beingproducedusingmanuallabour,industriessought machinesinordertoproducetheitems,leadingtothe Industrial Revolution in the 18thcentury. As of 2016, the worldenteredthefourthIndustrialRevolutioncoined as'Industry4.0',whichpromotesthecomputerisation trends: connectivity, intelligence, and flexible automation (AMFG AI 2019).

Industry4.0hasgivenrisetoanemergingsectorin manufacturingcalledSmartManufacturingthatopens doorsforanalyticsintheindustry.Figure1represents the various components attributed within the domain of Smart Manufacturing (Sodani 2021). It is a technology-drivenapproachthatutilisestheIoTandinternet-connecteddevicestoproducegoodsandmonitorpro-cesses.Itsgoalistoautomatetheprocessesinvolvedin manufacturing to maximise efficiency, increase sustain-ability,supplychainmanagement,andidentifythesys-tems barriers even before they occur by generating, opti-mising,andimplementingenormousvolumesofdata. With the application of advanced analytics to industrial data,manufacturerscangaininsightstooptimisethe productivity of individual assets as well as the total man-ufacturing operation prevailing artificial intelligence (AI) and machine learning (ML).

Industry4.0convergesinformationtechnologyand operational technology to create a cyber-physical ofmanufacturingbyencompassingthree technological

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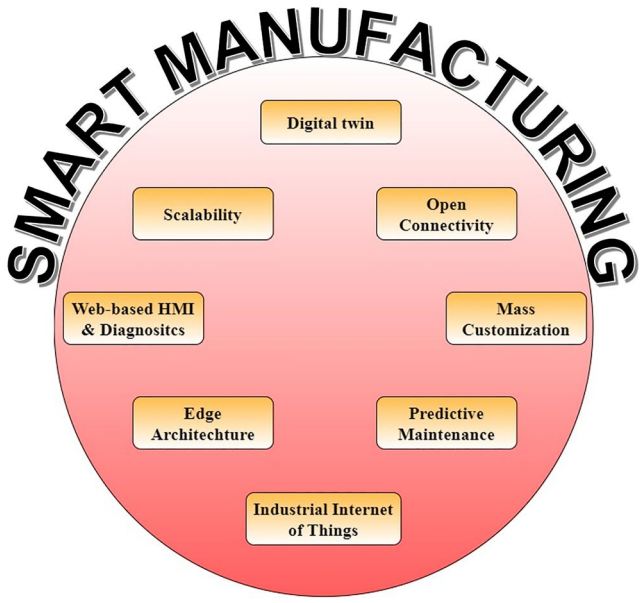

Figure1. Global Smart Manufacturing Segmentation.

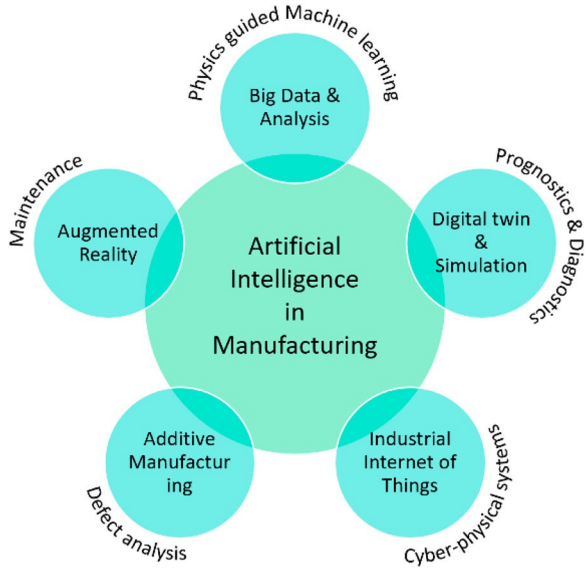

Figure2. The multi-faceted usage of AI and machine learning in manufacturing domain.

environmentwithdigitalsolutionsandadvancedtech-

subsetoftheInternetofThings(IoT)whichreferstoa networkofphysicaldevicesthataredigitallyintercon-nected, facilitating the communication and exchange of datathroughtheinternetIIoTinterconnectsmachines throughvarioussensors,RFIDtags,software,andelec-tronicswhichareintegratedwithmachinestocollect real-timedata.Theabundanceofsmartsensorsand theInternetofThingsarethekeyenablerofcurating andstoringaremarkableamountofindustrialdata-richenvironmentsrelatedtoallaspectsofproduction. Digitaltwinrunsanonlinesimulationbasedondata receivedfromIIoT.Adigitaltwinisadigitalrepre-sentationofareal-worldproduct,machine,process, orsystemthatallowscompaniestobetterunderstand, analyise and optimise their processes through real-time simulations.

Cloudcomputingusesinternetconnectionstostore, access, and process data and is a key enabler of the digital twinparadigm.Additionally,currentgenerationmanu-facturing is supported by advanced technologies such as advancedrobotics,augmented,andvirtualrealityand additivemanufacturing.MLtechniques,asubfieldof ArtificialIntelligence(AI),hasthepotentialtobecome the main driver in uncovering fine-grained intricate pro-duction patterns in smart manufacturing paradigm and offeringtimelydecisionsupportinawiderangeof manufacturingandproductionapplications,tonamea few,predictivemaintenance,processoptimisation,task scheduling, quality improvement, supply chain, and sus-tainability so on.

SmartuseoftechnologyandAIcanhelpgrowand drivebusinesses.Thebiggestcompaniesaroundthe worldareutilisingAIandMLinmanufacturingand investinghugeamountsofmoneyinitsdevelopment (Chuprina 2020). According to Mckinsey 40% of all the potentialvaluethatcanbecreatedbyanalyticstoday, comes from AI and ML techniques, wherein ML accounts for between $3.5 trillion to $5.8 trillion in annual value. Data-driventechniquesautomatelearningfromdata, detect prevalent underlying patterns and make informed decisions. Industries have been successful in the applica-tion of machine learning mainly into three aspects of the business:

• Industrial Internet of Things	• Additive	• Operations	• Productions	• Post-Productions
• Digital twin	• Advanced robotics			
• Cloud computing	• Augmented and virtual			

In manufacturing, a paradigm shift is happening right now. Advances in Big data and Machine Learning (ML) ischangingthetraditionalmanufacturingeraintothe

Additionally,dataanalyticsinthemanufacturing nologies including,

industry can be used to amplify growth in the following domains (but not limited to)

smartmanufacturin paradigmshiftiscre

(1)Improve assembly-line efficiency using data analytics.

(2)Improved Customer Experience, including curation, storage, personalisationandfindingindividualisedvalue propositions.

(3)Inventory management by

(a)Real-time insights and visibility into inventory **along the supply lnies**

(b)Delivery route optimizations

(4)Minimise loss associated with delayed, damaged, or lost goods in transport and for providing real-time asset management include real-time alerts

(5)Reduce errors and corrections during product development and improve the product's quality and pack-

enable real-time actionable intelligence to advanced top-ics such as edge computing, fog computing, cybersecurity aspects of smart manufacturing. *Hence, this special issue is focused on bringing together a wide range of researchers to report the latest efforts in the fundamental theoretical as well as experimental aspects of ML and their applications in manufacturing and production systems*.

Summary of papers in this special issue

Asummaryofpapersthataddressseveralimportant issuesattheintersectionofmanufacturingandML

aging with,

(a)Analytics-backed simulations

(b)Product modelling

(6)Predictivemaintenancehelpsincreaseassets’life-

domain is presented next. Different categories to which **the paeprs belong and the importance of these categories** inmachinelearninginthemanufacturingcontextare elaborated next.

time by,

(a)Asset Management.

(b)Improving asset availability.

(c)Detection of faults and defects.

(d)Prevention of unplanned downtimes.

(7)Increase visibility into the supply chain with action-able data location-based IoT services.

The manufacturing environment offers enormous oppor-tunitiestoleverageartificialintelligencetechniquesfor better decision-making (Sharma, Zhang, and Rai 2021). AkeyenablerofAIinmanufacturingenvironmentsis IIoT,whichfacilitatedtheharnessingofbigdata.Big data,inturn,needsAItomakedecisionsinreal-time. The core techniques of AI like deep learning, computer vision, reinforcement learning have already been utilised inmanufacturingscenarios.Computervisionhasbeen leveragedforstructuralhealthmonitoring(Yangetal. 2020).Thevision-basedtechniquesofferedhighspatial resolution even with inexpensive sensors. Ontology can be used to store and organise the knowledge of manufac-turing systems (Ali et al. 2019). AI can be used to study thebraincomputerinterfaceofmanufacturingopera-tors (Shankar and Rai 2014). This can enhance the study of industrial safety and ergonomics. The data generated by manufacturing processes, like additive manufacturing, canbeanalyisedforreal-timemonitoring,control,and defectminimisation(Khadilkar,Wang,andRai2019). Themanufacturingequipmentcanbediagnosedand prognosed using machine learning (Yang and Rai 2019). Augmentedrealitydeviceshavebeenusedformainte-nanceapplications(YoungandRai2021).Theyoffered additional flexibility to reconfigure the maintenance pro-

Computer vision-based inspection and monitoring

One of machine learning's most high-impact application areas in the manufacturingdomain is computer vision-based part inspection and process monitoring. Utilising cost-effective sensors such as RGB cameras wrapped with ML-basedalgorithmscanenablehighthroughputpart inspection.Computervision(imagesandvideo)based approaches that are integrated with ML can enable moni-toring of a product throughout the entire production pro-cess. Additionally, a computer vision-based approach can also enable high-quality continuous process monitoring.

'Adata-drivenmethodforEnhancingtheimage-basedautomaticinspectionofICwirebondingdefects'byChenetal. (Chen,Zhang,andWu2020)offersa data-drivenmethodforenablingautomaticinspection of integrated circuit (IC) wire bonding defects. The out-lined methodin the papercomprisesof three steps:(1) datapre-processingforlocatingandseparatingICchip image patches from the raw image, (2) feature engineer-ing that extracts geometric features from the segmented wires, and (3) machine learning algorithms (such as CNN and SVM) based classification. The authors showcase the efficacy of the developed method on a set of X-ray images collected from a semiconductor factory.

'Softsensorofflotationfrothgradeclassification basedonhybriddeepneuralnetwork'byZhangand Gao(ZhangandGao2021)summarisestheireffortsto developasoftsensorthatcanprocessfrothimagesof flotation tailing to classify iron ore tailings grade. The two key contributions of the paper are to curate and establish

cedure on the go. a database of froth images of flotation tailings and a com-

WhiledifferentMLtechniqueshavebeenusedina variety of manufacturing applications in the past, many openquestionsandchallengesremain,fromBigdata

parison of the accuracy of multiple deep neural network (DNN) models for the task at hand. Based on the comparisonresults,theauthorsoutlineafine-tunedhybrid

DNNmodelthathasexcellentaccuracyanddevelops softwarearoundit.Theexperimentalresultsshowthe potential of application of DNNs in the field of iron ore

machine learning-based manufacturing solutions at scale will be key in the near future.

‘Acloudedge-basedtwo-levelhybridscheduling froth flotation. learningmodelincloudmanufacturing’byJianetal. (Jian, Ping, and Zhang 2020) presents a cloud edge-based two-level hybrid scheduling learning model to enable fast prediction of the scheduling results. The proposed model

Fault detection

Timely and accurate diagnosis of manufacturing equipment process faults provides a strategic advantage to help manufacturingcompaniesstaycompetitivebyreducingmachinedowntimes.Machinelearningalgorithms withinthemanufacturingindustrywillfindincreased usageforenablingmanufacturingsystemfaultdiagnosisasmoreandmorecustomersrequiremanufacturers to expedite the delivery of high-quality products at a low cost.

‘A deformable CNN-DLSTM based transfer learning method for fault diagnosis of rolling bearing under multi-ple working conditions’ by Wang et al. (Wang et al. 2020) presentsatransferlearning-basedapproachthatcombinesdeformableconvolutionalneuralnetwork(CNN) anddeeplongshort-termmemory(DLSTM)rolling bearing fault diagnosis. The authors specifically focus on bearing faults under multiple working conditions where it is hard to find sufficiently large-scale labelled data. The transfer learning approach allows them to pre-train fault diagnosis model using data samples under one working condition and then transfer the model with further fine-tuning (with very few data samples) to another working condition.Thedevelopedframework,whenappliedto anexperimentaldataset,showsbetterresultsthanthe state-of-the-art methods.

‘Applicationsofdeeplearningforfaultdetectionin industrialcoldforging’byGlaeseretal.(Glaeseretal. 2021) showcase the usage of deep learning techniques in the industrial cold forging fault detection domain. Specifically, the authors outline two different approaches based onconvolutionsneuralnetwork(CNN)anddecision tree(DT)tounderstandhowmachinesignalimpacted variousfaults.Theoutlinedapproachesshowsuperior performance on vibration test data sets collected for com-monlyencounteredfaultsintheindustrialcoldforging domain.

Cloud manufacturing

Cloudstoragewithinmanufacturingisbeingembraced to store and manage manufacturing production big data. Aside from the understandably increased storage space, thecloudandmachinelearningtechniqueswillhelp reduce costs and drive profitability. Understanding how hardware–EdgevsCloud–indeployingreal-world

atthefirstleveldecomposesanddistributesmultiple schedulingtasksintoseveralfirst-levelsub-tasksbased onFirst-in First-out(FIFO)principle.Thesecondlevel enables even granular decomposition to atomic tasks that are then scheduled to different industrial devices (factory edge nodes). The second level utilises an innovative combinationofimprovedbatalgorithm(VSSBA)andlong and short-term memory networks (LSTM) for schedulingproblems.Experimentsperformedbytheauthors showthatthetwo-levelapproachcanimproveperfor-mance in real-life applications in a cloud manufacturing setting.

Process improvement and optimisation

TheprescriptiveanalyticspowerofMLtechniquescan augmentthemanufacturingemployees'efforttoselect theoptimalsetofparametersrelatedtoagivenman-ufacturingprocess,henceenablingmanufacturingpro-cess improvement and optimisation. The discipline at the intersection of ML and process improvement that leads to manufacturing analytics insights enabling faster mass and customised production at a rapid pace with as little waste as possible will grow by leaps and bound in the next decade.

'AninvestigationoftheutilisationofdifferentdataSourcesinmanufacturingwithapplicationinInjection Moulding' by Rønsch et al. (Rønsch, Kulahci, and Dyb-dahl 2021) report their work on the study that explores effectiveutilisationofvariousdatasourcesforprocess improvementininjectionmoulding.Specifically,the authorsinvestigatewhetheragoodpredictionaccuracy can be achieved by using readily available Machine Pro-cess Data on a moulding manufacturing line comprising of 100 injection moulding machines or additional sensor signals obtained at a higher cost can provide additional beneficial information. For the specific use case that was carried out in close collaboration with an industrial part-ner,theyconcludethatavailablemachineprocessdata doesnotcapturethevariationintherawmaterialthat impacts element quality.

'Usingprocessminingtoimproveproductivityin make-to-stockmanufacturing'byLorenzetal.(Lorenz etal.2021)demonstrateanovelusecaseofdata-drivenproceduretoimproveproductivityinmake-to-stock manufacturing. Specifically, the outlined procedure

utilises process mining to dynamically map and analyse manufacturingprocesseswithhighprocesscomplexity andvarietyinanautomatedmannertoenableproduc-tivity improvement. The authors also empirically validate the usage of the developed procedure on a test case emanating from a leading manufacturer of sanitary products andprovidetangibleimprovementsuggestionsforthe

acomprehensivehybridliteraturereviewapproach comprising of a thematic and conceptual synthesis of the literature resulting in the interpretive model framework ofmanufacturing.Theresultantinterpretativeframe-work is articulated as consisting of scan, store, interpret, execute,andlearnasitspurposivecomponents.Additionally,theauthorsprovidepertinentfutureresearch

manufacturers. questionsandimplicationsformanufacturingopera-

‘Asupervisedmachinelearningapproachforthe

optimisationoftheassemblylinefeedingmodeselec-

tions, manufacturing strategy, and manufacturing.

tion'byZangaroetal.(Zangaro,Minner,andBattini 2020)outlinesasupervisedlearning-basedapproach thatutilisesclassificationandregressiontree(CART) algorithm for line feeding problem (LFP). The proposed approachtakesasinputattributesofcomponentsand manufacturing environment to create a decision tree that suggests a line feeding mode for every component. Addi-tionally,forcasesthatresultinaninfeasiblesolution, theyalsooutlinearepairapproachthatprovidesfeasi-ble solutions with respectable average cost deviation. The proposed approach predicts line feeding mode with good classification accuracy.

State of the art review papers

Reviewpapersrelatedtomachinelearningapplications in the manufacturing domain in this special issue bring togetherquantitativeandqualitativecomponentsand providenewconceptualframeworks,synthesisediverse

Conclusion

results, and give the broader research community a 'state-of-the-art' snapshot of essential issues related to the focus of this special issue.

'Artificial intelligence (AI) in augmented reality (AR)-assistedmanufacturingapplications:areview'bySahu etal. (Sahu,Young,andRai2020)providesacompre-hensivereviewofaverydynamicresearchdomainat theintersectionofAIandAR.Thecurrentparadigms utilisedinARforcameracalibration,detection,track-ing,camerapositionandorientation(pose)estimation, inverse rendering, procedure storage, virtual object cre-ation, registration, andrendering arestill mostlydomi-nated by traditional non-AI approaches. The authors pro-vide a critical appraisal of these strategies. Additionally, the authors also outline potential AI solutions for every component of the computational pipeline. Furthermore,

Manufacturingisundertakingadefinitiveshiftasit assimilatesandistransformedbytheusageofmachine learning techniques. This special issue brought together awiderangeofresearcherstoreportthelatestefforts inthefundamentaltheoreticalaswellasexperimental aspectsofMLandtheirapplicationsinmanufacturing and production systems to understand and contextualise thisshiftattheonsetofthisdecade.Thetopicssuch ascomputer-visionbasedinspectionandmonitoring, faultdetection,cloudmanufacturing,processimprove-mentandoptimisation,andcomprehensivestateofthe art review papers are spanned by the papers in this special issue.

Goingforward,asIndustry4.0transformstheman-ufacturingenvironmentsintocyber-physicalenviron-ments,AIwillbeindispensablefordataanalysisand subsequent decision-making (Rai and Sahu 2020). Prod-uct design (Huang and Rai 2018), inspection (Zhang et al. 2019a),geometryprocessing(Zhang,Jaiswal,andRai 2018),materialinformatics(Wangetal.2021),control ofdynamicsystems(Zhangetal.2021)arefewmore applications of ML in manufacturing. The utility of ML inmanufacturingapplicationsreflectsthatMLcanbe integrated into every phase of product lifecycle, starting from design to disposal of the product in a manufacturing environment.

ItisalsoimportanttonotethattheML-based decision-making in manufacturing applications can also besupplementedbythephysicsgoverningthephysical phenomena. Synergizing ML with physics becomes cru-cial, especially in the case of manufacturing applications where data generation can be expensive and unsafe (Rai and Sahu 2020).

interesting and fruitful future research work directions at the intersection of AI and AR are also listed.

'The interpretive model of manufacturing: A theoret-ical framework and research agenda for machine learn-ing in manufacturing' by Sharma et al. (Sharma, Zhang, andRai2021)arguesthatanewframeworkisneeded tofullycomprehendtheparadigmofapplicationof machine learning in manufacturing. The authors present

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